

Jacob Kang

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Research Interest

- Computational Neuroscience
- Brain Imaging
- Machine Learning
- Nonstationary Dynamics

Education

University of Maryland, College Park, Ph.D. Student in Bioengineering Aug 2026 –

- Doctoral Advisor: Prof. Nan Xu
- Relevant Coursework: Network Neuroscience and Dynamic Analysis, Machine Learning and Statistical Signal Processing, Random Processes in Communication and Control

University of Maryland, College Park, B.S. in Computer Science and Mathematics: Statistics Jan 2024 – May 2026

- GPA: 3.91/4.00; Graduated with **Cum Laude**
- Academic Mentor: Prof. James A. Yorke
- Relevant Coursework: Computer Vision, Bioinformatics Algorithm, Numerical Analysis, Statistical Computing with SAS, Sampling Theory, Application of Linear Algebra, Advanced Data Structure, Advanced Calculus, Differential Equations

Pennsylvania State University, University Park, Studied Computational Data Science Aug 2022 – Dec 2023

- GPA: 3.91/4.00
- Relevant Coursework: Data Science through Statistical Reasoning, Data Management for Data Science, Programming Models for Big Data, Probability Theory

Publications

A Conserved Neural–BOLD Brain State Reveals Glia-Associated Neurovascular Coupling Jun 2026

Linnea Hoheisel †, **Jacob Kang** †, Archana Sathiyamoorthy, Francesca Mandino, Edward Bernat, Evelyn Lake, Nan Xu
Co-first author; In preparation for *Nature Neuroscience*

Beyond Gaussian Assumptions: Distribution-Aware Channel Capacity for Brain Effective Connectivity Jun 2026

Jianan Jian, Nurahmed Multezem, Benjamin Li, **Jacob Kang**, Nan Xu
Co-author; Submitted to *NeurIPS*

Clustered pattern projection for EEG dementia classification: evaluating the generalizability of disorder patterns Jun 2026

Hunseok Kang †, **Jacob Kang** †, Mustafa Zeki, Jonghyeon Seo
Co-first author; Accepted to *Frontiers in Neuroscience*

Dynamic mode decomposition–based clustered pattern projection for reliable Alzheimer’s disease detection from EEG Feb 2026

Jonghyeon Seo, Hunseok Kang, **Jacob Kang**, Aymen I. Zreikat
Co-author in *Diagnostics* doi.org/10.3390/diagnostics16040530

CNN-based framework for Alzheimer’s disease detection from EEG via dynamic mode decomposition Nov 2025

Jacob Kang, Hunseok Kang, Jonghyeon Seo
First author in *Frontiers in Neuroinformatics* doi.org/10.3389/fninf.2025.1706099

† These authors have contributed equally to this work and share first authorship

Presentations

A reproducible neural–BOLD state motif links brain-state dynamics across modalities and species Nov 2026

Jacob Kang †, Linnea Hoheisel †, Archana Sathiyamoorthy, Francesca Mandino, Edward Bernat, Evelyn Lake, Nan Xu
Submitted to *BRAIN Initiative 2026*

Linnea Hoheisel †, **Jacob Kang** †, Archana Sathiyamoorthy, Francesca Mandino, Edward Bernat, Evelyn Lake, Nan Xu
Submitted to *Society for Neuroscience 2026*

Brain states in simultaneous EEG-fMRI

Jun 2026

Linnea Hoheisel †, **Jacob Kang** †, Archana Sathiyamoorthy, Edward Bernat, Nan Xu
Organization for Human Brain Mapping 2026

Temporal coupling of neural and BOLD dynamic states from simultaneous resting-state EEG-fMRI

Apr 2026

Jacob Kang, Linnea Hoheisel, Nan Xu

Mid-Atlantic Research Day at UMD/JHU & Undergraduate Research Day at UMD

Poster
Extended Abstract

Modeling the dynamics of breakthrough infections and vaccination stagnation

Apr 2026

Jacob Kang, Kejia Zhang, Joshua Weitz

Undergraduate Research Day at UMD

Poster

Research Experiences

Undergraduate Researcher, INSPIRE Lab

Nov 2025 – May 2026

University of Maryland, College Park

Mentored by Prof. Nan Xu, Linnea Hoheisel (Postdoctoral Researcher)

- Found transient cross-frequency neural-state co-expression pattern is significantly often co-occurring with BOLD state showing prominent negative engagement of the default mode and frontoparietal network around onset of the co-expression
- Validated with two independent simultaneous EEG-fMRI data in humans ($n = 22$ and 12 ; $TR = 2$ s and 2.52 s) and calcium fMRI data in mice and observed reproducible pattern across modalities and species
- Developed Hidden Markov Model-based brain state analysis framework that examines recurring brain states from neural, cellular, hemodynamic signals and may support future biomarker studies in disorders involving frontoparietal dysfunction (e.g. ADHD), enabling more accessible investigation of large-scale brain dynamics

REU Student & Undergraduate Researcher, Weitz Group

Jun 2025 – Jun 2026

University of Maryland, College Park

Mentored by Prof. Joshua Weitz, Kejia Zhang (PhD Student)

github.com/mdsnpr/Epidemiology_UMD

- Developed SIRV epidemic model implementing both deterministic and stochastic simulations to evaluate vaccination strategies across parameter space of R_0 and vaccination rate (v)
- Identified critical vaccination thresholds and optimal coverage strategies that maximize epidemic control while minimizing resource allocation; presented findings at UMD Summer REU Poster Session
- Extended model to incorporate vaccine failure rates, revealing a counterintuitive threshold: below critical failure rates, peak breakthrough infections occur at intermediate (not maximal) vaccination coverage

Lead Researcher, Independent Research

May 2025 – Oct 2025

Mentored by Dr. Jonghyeon Seo

github.com/mdsnpr/AlzDetect_EEG_DMD

- Developed Alzheimer's classification framework using EEG data from 63 subjects (36 patients, 27 controls), combining Dynamic Mode Decomposition with Sequential CNN to achieve 74.23% accuracy in Leave-One-Subject-Out validation, outperforming recent baseline (62.50%) and FFT-based methods (70.56%)
- Adopted stacked shortened window (2 seconds * 12) approach when forming input to Sequential CNN to preserve nonstationary dynamics, especially emphasized in eyes open EEG data
- Identified delta (δ) frequency band as a confounding factor in Alzheimer's classification; excluding δ resulted in statistically significant improvement in accuracy by 1.27% ($p < 0.05$, paired t-test)

Work and Teaching Experiences

Supervisor, Nebula Research Data Fellowship, University of Maryland

Sep 2026 –

Faculty Assistant, INSPIRE Lab, University of Maryland

Jun 2026 –

- Conducting computational neuroscience research and in-lab IT support for INSPIRE LAB under Prof. Nan Xu

Grader for Differential Equations, University of Maryland

Sep 2025 – Dec 2025

Learning Assistant for Introduction to Data Science, Pennsylvania State University

Aug 2023 – Dec 2023

Advising and Mentoring Experiences

Vishal Senthilkumar, Undergraduate student in Computer Science, University of Maryland

May 2026 –

- Advising on computational neuroscience research in Quasi-Periodic Pattern-based Biomarker study using fMRI from Infants

with Prenatal Opioid Exposure

Awards and Honors

CMNS Scholarship Funds (\$3000), University of Maryland	2025
President’s Transfer Scholarship (\$5000), University of Maryland	2024
President Walker Award, Pennsylvania State University	2023
Top Prize in Big Data Utilized Idea Competition, Seoul Metropolitan Office of Education	2021
Medal and Certificate in Euclid Mathematics Competition, University of Waterloo	2021

Skills

Programming: MATLAB, Python, R, SAS

Computational Methods: Hidden Markov Models, Dynamic Mode Decomposition, Machine Learning, Time-Series Analysis, Mathematical and Statistical Modeling

References

James A. Yorke Department of Mathematics and Physics, University of Maryland, College Park <i>Academic Mentor</i>	yorke@umd.edu
Nan Xu Fischell Department of Bioengineering, University of Maryland, College Park <i>Doctoral Advisor</i>	nanxu@umd.edu inspirelab.umd.edu
Joshua S. Weitz Department of Biology, University of Maryland, College Park <i>Undergraduate Research Advisor</i>	jsweitz@umd.edu weitzgroup.umd.edu
Lawrence C. Washington Department of Mathematics, University of Maryland, College Park <i>Coordinator of REU program in Infectious Disease Dynamics</i>	lcw@umd.edu